



Defining features of linear relationships

Task A: Numerical features of linear relationships

1) These are tables of values for linear relationships. Fill in the missing values and complete the statements about the **rate of change**.

a)

x	-2	-1	0	1	2	3	4
y	-8	-3	2	7	12	17	22

For every change of +1 in x there is a change of +5 in y

b)

x	-2	-1	0	1	2	3	4
y	-18	-11	-4	3	10	17	24

For every change of +1 in x there is a change of +7 in y

c)

x	-2	-1	0	1	2	3	4
y	-1	0.5	2	3.5	5	6.5	8

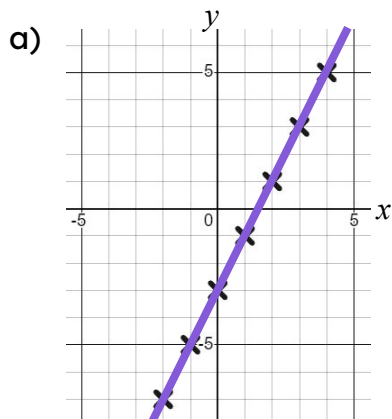
For every change of +1 in x there is a change of +1.5 in y

d)

x	-2	-1	0	1	2	3	4
y	22	17	12	7	2	-3	-8

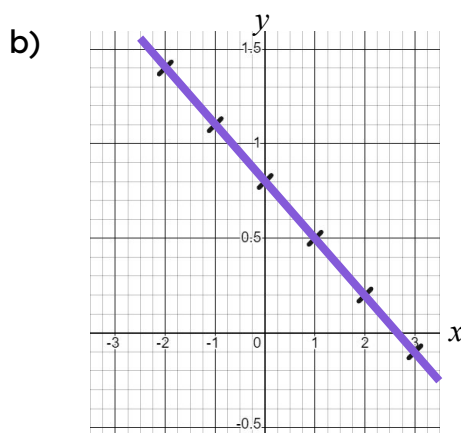
For every change of +1 in x there is a change of -5 in y

2) Use the graph to complete the table and write a sentence about the **rate of change**.



x	-2	-1	0	1	2	3	4
y	-7	-5	-3	-1	1	3	5

For every change of +1 in x there is a change of +2 in y



x	-2	-1	0	1	2	3
y	1.4	1.1	0.8	0.5	0.2	-0.1

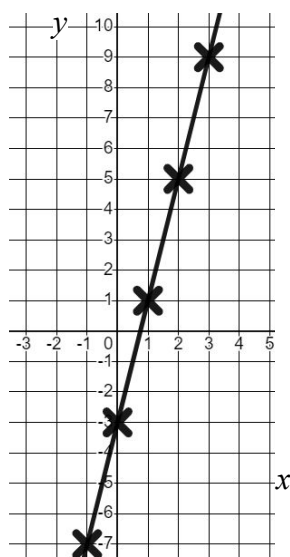
For every change of +1 in x there is a change of -0.3 in y

Task B: Graphical features of linear relationships

1) Izzy has completed a table of values and plotted coordinates for the linear relationship $y = 4x - 3$

x	-2	-1	0	1	2	3	4
y	-11	-7	-3	1	5	9	13

The line $y = 4x - 3$ starts at 1 and has a **rate of change** of +4 each time.

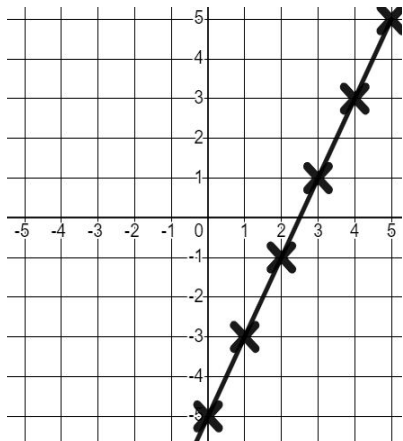


The line $y = 4x - 3$ has no start or end. You need to continue the line in both directions and plot zero and negative x values.

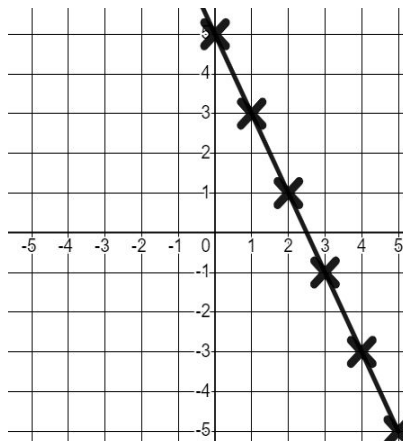


2) Match the graphs to their unique descriptions.

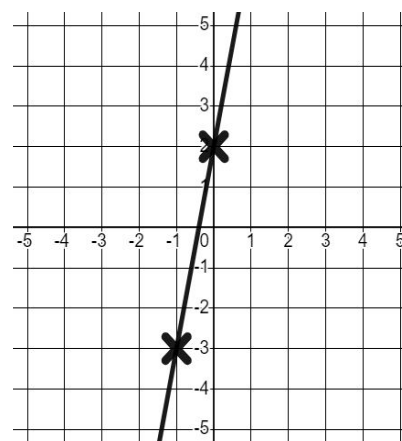
Intercepts the y -axis at $(0, -5)$, rate of change of $+2$



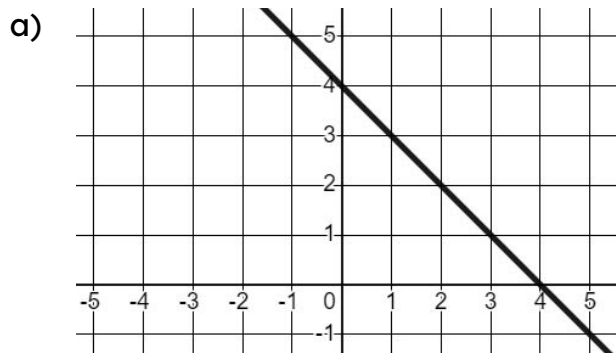
Intercepts the y -axis at $(0, 5)$, rate of change of -2



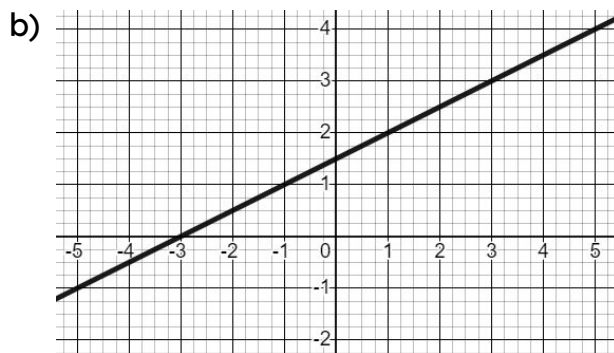
Intercepts the y -axis at $(0, 2)$, rate of change of $+5$



3) Write unique descriptions for these two graphs. Be sure to include the words y -intercept and rate of change..



Intercepts the y -axis at $(0, 4)$, rate of change of -1



Intercepts the y -axis at $(0, 1.5)$, rate of change of $+0.5$

